

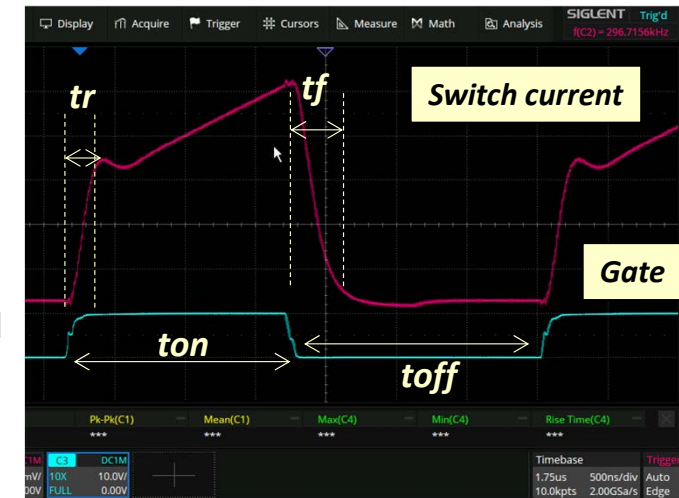


Input Capacitor Sizing and Selection in DC/DC Power Supplies

Power Supply Hardware Design Down to Component Level
(Face-to-Face and Online Workshop)

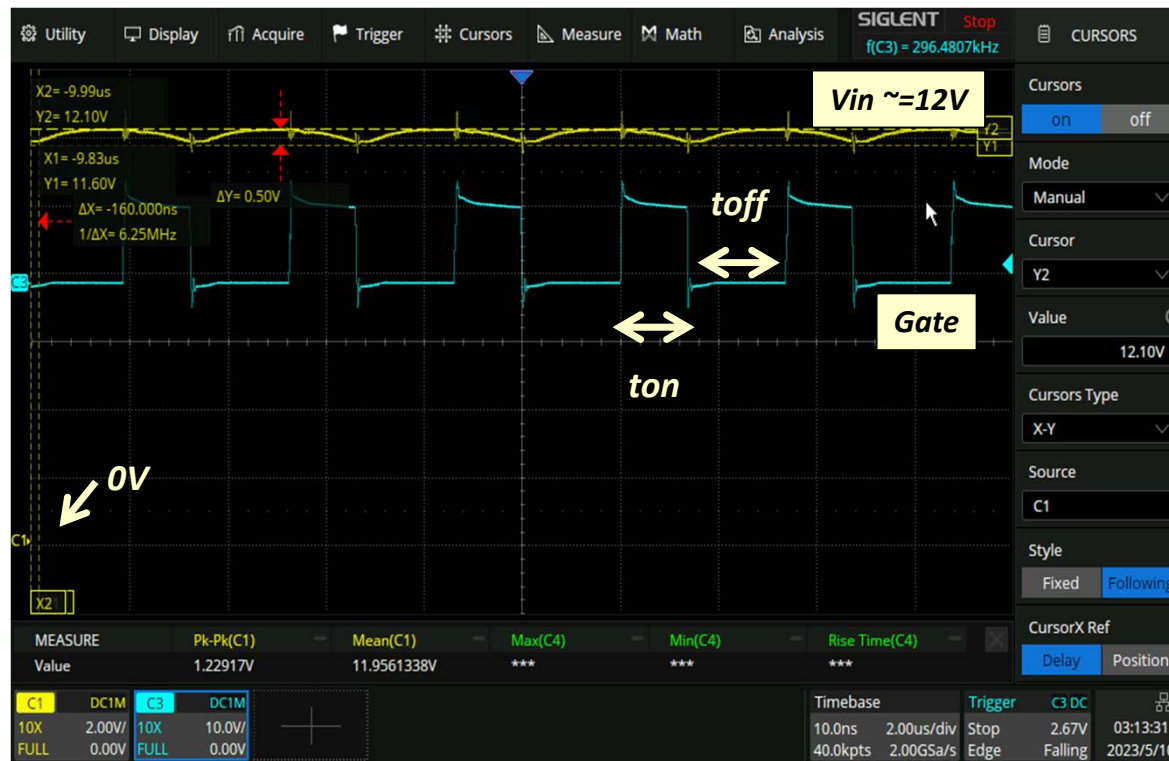
Why Do We Need Input Capacitance?

- There are 2 main reasons: Noise Reduction & Energy Storage
- Noise Reduction → usually the job of ceramics
 - To reduce voltage ripple on the input voltage (V_{in}) during our transistor's ON period (t_{on}) and OFF period (t_{off})
 - During t_{on} , we draw current from V_{in} therefore it falls
 - During t_{off} , main DC Bus voltage charges the input caps on V_{in} → V_{in} rises
 - Therefore we get a ripple on our V_{in} → more on this later
 - To reduce the spikes as a result of switch's fast rise time (t_r) and fall times (t_f)
 - That is because di/dt during these two period is huge → again more on this soon
- Energy Storage/Bulk capacitance → usually the job of electrolytics
 - To reduce slow voltages dip after a big load step
 - By slow we mean relative to our switching frequency and switch transition
 - If we suddenly turn on a massive load, the needed energy has to come from somewhere
 - To “hold-up” V_{in} above our minimum threshold, if our main DC Bus voltage is interrupted for a little while → called Holdup Time(t_{hold})
 - We can calculate the of size our bulk caps to store enough energy so that V_{in} does not fall below a specified level for a specified “holdup” time (t_{hold})
 - Again this slow transient is usually the job of our electrolytic Bulk Caps



Reducing Voltage Ripple on Vin

- You can see that during t_{on} , as we draw current V_{in} dips; during t_{off} , our DC Bus charges the input caps and V_{in} rises
 - This gives us a peak to peak ripple voltage ($\Delta V_{in_pk_pk}$), in this case $\sim 500\text{mV}$ *



* We have really exaggerated this for teaching purposes

Reducing Spikes on Vin During Rise and Fall Times

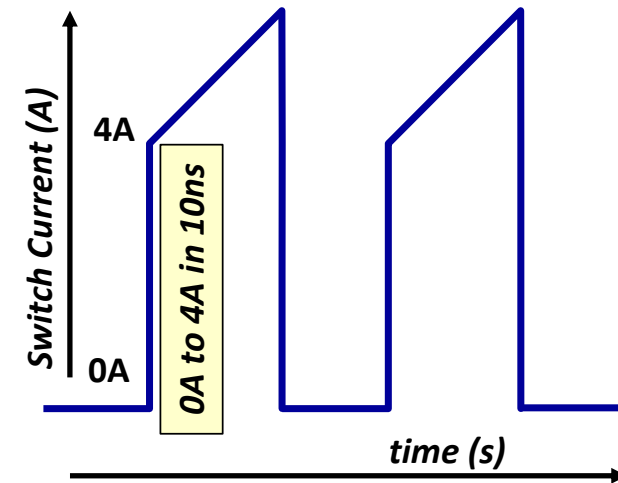
- You can see the switching spikes* on Vin (yellow trace) occurring during the rise and fall times of the switch current (pink trace)
 - To remove these, we need capacitors that work well at high frequencies → ceramics are a good choice → more on this later!



* We have really exaggerated this for teaching purposes

Why Do We Get Spikes During Switching Transitions?

- Many switching power supply topologies (e.g. Buck/Flyback) draw very choppy current with fast rise/fall times as shown
- Let us assume that at time $t = 0s$ our switch current is $0A$ and when we turn on the switch of our power supply, it rises to $4A$ in $10ns$
- Let us also assume that our switching frequency $F_s = 200kHz$
 - Therefore, turn on happens every period $T = 5\mu s$ and our rise time $t_r = 10ns$
- Let us also assume that our PCB track is $4cm$ long i.e. it has ~around $40nH$ of inductance (L)
 - A very rough rule of thumb is that $1cm$ of PCB track has $\sim 10nH$ of inductance
- We know from school level Physics that for an inductor: $E = -L \frac{di}{dt}$
 - Where E is the Back EMF generated across the inductor in volts, L is our inductance and di/dt is the rate of change of current
 - So if our current is going from 0 to $4A$ in $10ns$ then our $di/dt = 4/10ns \rightarrow$ i.e. $400 \times 10^6 A/s!$
 - Our PCB track has an L of $40nH$, so from the equation above we will have a voltage spike of $40nH \times 400 \times 10^6 = 16V$ happening once every rise time i.e. once per switching cycle
 - We will have another spike every fall time $\rightarrow$ fall time spike is usually bigger
 - Google “Biricha YouTube Inductors” for our video on how inductors act in PSUs



We need some capacitance to dampen these spikes otherwise our entire power trace may become noisy

Reducing Voltage Dips if There Is a Big Load Step

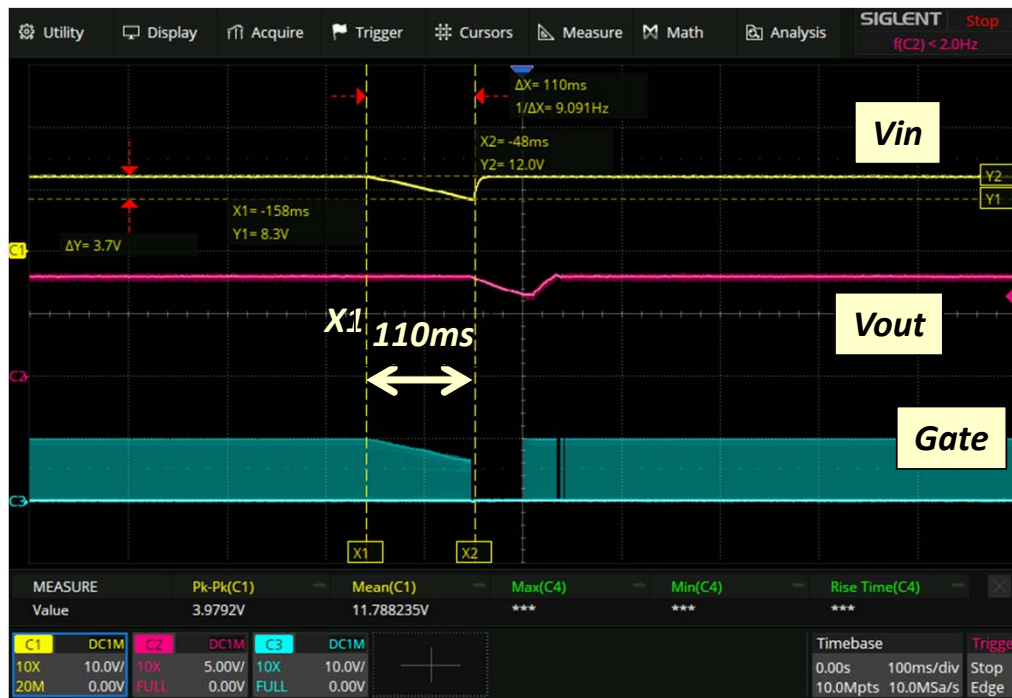
- The pink trace shows the moment that we give our PSU a massive load step
- The yellow trace is our V_{in} without sufficient Bulk Capacitance* → it dips!
 - The more Bulk Capacitance we have, the smaller the dip → usually electrolytics



* We have really exaggerated this for teaching purposes

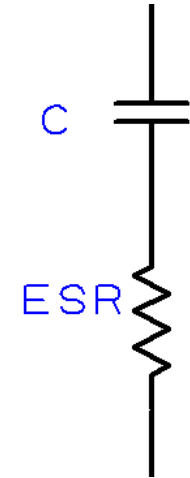
Holding Vin above Minimum Threshold, if Line Voltage Is Interrupted

- At $t = X1$, we cut the power and our Vin (yellow trace) falls
 - Our bulk caps slow down the fall $\rightarrow$ after 110ms, Vin it has fallen from 12V down to 8.3V
 - The bigger our input capacitance (C_{in}), the slower this fall



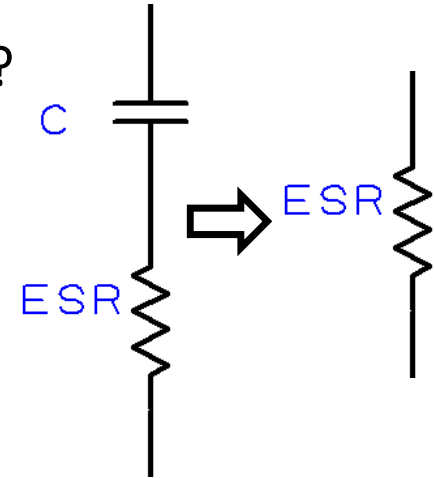
What Type of Capacitors Do We Use As Input Capacitance?

- For reducing ripple and spikes and noise removal, we normally use Ceramics
 - This is because ceramics work better at high frequencies than electrolytics
 - Turn on/off times are typically in the order of a few micro-seconds
 - Rise and fall times are in the order of 10s of nano-seconds
 - Electrolytics are not the best choice → more on this soon
- For load steps and DC Bus interruptions/Brownouts we usually use electrolytics
 - This is because we need lots of capacitance as bulk energy storage → electrolytics are cheap for large uF values
 - Furthermore, load steps and brownouts are slow events when compared to tr/tf etc. and electrolytics work very well at low frequencies
- Electrolytics are not suitable for noise removal/high frequency events, due to their large Equivalent Series Resistance (ESR)
 - By contrast the ESR of ceramics is very low and therefore when we talk about ESR in the context of PSUs, we are often talking about the electrolytics
 - Aluminium polymer and Film caps however are very good ! Please Google “YouTube Biricha capacitor” to find our videos on this topic
 - We can parallel identical electrolytics to reduce their combined ESR but this will take a lot of space on the PCB



Why Capacitors with Large ESR Are Not Good at Reducing Voltage Ripple?

- As our frequency goes up, the capacitive impedance of our capacitor falls but its parasitics ESR remains “relatively” constant
- At high frequencies, our capacitor’s capacitive impedance becomes so low that it becomes very small compared to the ESR and current will only see the ESR
 - Because $1/(2\pi fC)$ has become so small it is tiny compared to the ESR
- Ripple current through our ESR will then produce a ripple voltage proportional to the ESR and cause resistive losses as per Ohms Law
 - Ceramics have very low ESR so we almost always ignore these losses → often ripple current rating of ceramics is not even specified by the manufacturer!
 - Electrolytics have much bigger ESR and the manufacturer states the maximum amount of ripple current for them so as to limit overheating
 - This is why we often use our ceramics to bring down the ripple voltage, and hence divert the ripple current that would otherwise go through the electrolytic and cause overheating in its ESR
- Therefore, for effective voltage ripple reduction and noise removal, you need to select a type of capacitor that has very low ESR
 - Ceramic/Film/Aluminium Polymer
 - Several equal value electrolytics in parallel to divide down the ESR
 - Please Google search “Biricha YouTube Capacitor” for our videos on this subject



How to Select Our Input Capacitance?

- Based in the previous discussions, we typically need
 - A bunch of ceramics in order to reduce the ripple voltage and the spikes
 - This has the added advantage that it will also reduce the ripple current through the electrolytics
 - One or more electrolytics as bulk capacitance so that we do not get a voltage dip after a load step or after a brownout*
- The question then is how do we size the amount of our ceramics and electrolytics
- In order to size our ceramics we need to know the amount of our input ripple current without any filtering
 - We will talk about how we do this shortly
- In order to size our electrolytics we usually need to know how much input voltage dip we can tolerate
 - We will talk about all these issues shortly
 - For small powers, we may be able to get away with not adding bulk capacitance if some is provided close by elsewhere on the board, but if you have enough space/money, it is always prudent to put some in

How to Size the Amount of Ceramics?

- Our specification usually determines our maximum Peak to Peak Input Voltage Ripple $\Delta V_{in_pk_pk}$
 - $\Delta V_{in_pk_pk}$ of $\sim 100\text{mV}$ or $\sim 1\%$ of V_{in} (whichever is smaller) are used often for V_{in} of up to 60V , but there is no real rule of thumb
- The biggest problem is that, unless you are following a specific Standard or are using a LISN, often the impedance of the source that is supplying your power supply is unknown
 - We therefore have to design for a worst case scenario and assume that we have a very poor high impedance energy source feeding our power supply
 - This is not an unreasonable assumption if our power supply is far away from the DC Bus + it is prudent $\rightarrow$ we will make assumption for the rest of this presentation
- The easiest way to determine the amount of Input Capacitance (C_{in}) that you need in advance is to simply simulate your power supply and then add enough capacitance until you meet the $\Delta V_{in_pk_pk}$ requirements
 - Or you can just simulate a current source with the correct shape and amplitude
- Alternatively Biricha's Automated Power Supply Design Software i.e. Biricha WDS, automatically calculates the amount of C_{in} that you need for a given desired $\Delta V_{in_pk_pk}$
- If you do not have access to the either Biricha WDS or a simulator, then there are standard equations that relate $\Delta V_{in_pk_pk}$ to C_{in} in term of quantities that we know
 - Please see Mankitala ISBN 978-0-12-386533-5 but we will give a full worked example shortly

How to Size the Amount of Ceramics Worked Example

- Let us assume we have a Buck converter with the following specification:
 - $V_{in} = 12V$ (Nominal Voltage V_{in_nom}), Output Voltage $V_{out} = 3.3V$, Output Current $I_o = 2A$, Switching Frequency $F_s = 200kHz$ and Duty $D = 32\%$
 - Max acceptable Input Ripple Voltage $\Delta V_{in_pk_pk} = 100mV$
 - $V_{in_max} = 14V$ (Maximum rail voltage), $V_{in_min} = 12V$ (Minimum rail voltage)
- If you have designed your power supply using WDS then it will have automatically calculated the amount of C_{in} that you need
 - It will also calculate all the necessary stresses such as maximum voltage stress
 - Please note that WDS assumes that the source Z is unknown and therefore designs for a worst case scenario → i.e. High Z source
- If you do not have WDS then you can simulate it
- If you neither have WDS nor a simulator, you can use the standard equations in the literature*
 - Again please note that these equations assume high source impedance
 - Example on the next slide

- * See Mankitala ISBN 978-0-12-386533-5, for most popular topologies – pp. 711 & pp. 176

How to Size the Amount of Ceramics Worked Example

- For example $\Delta V_{in_pk_pk}$ for a Buck converter is:

$$\Delta V_{in_pk_pk} \approx \frac{I_o D(1 - D)}{F_s C_{in}}$$

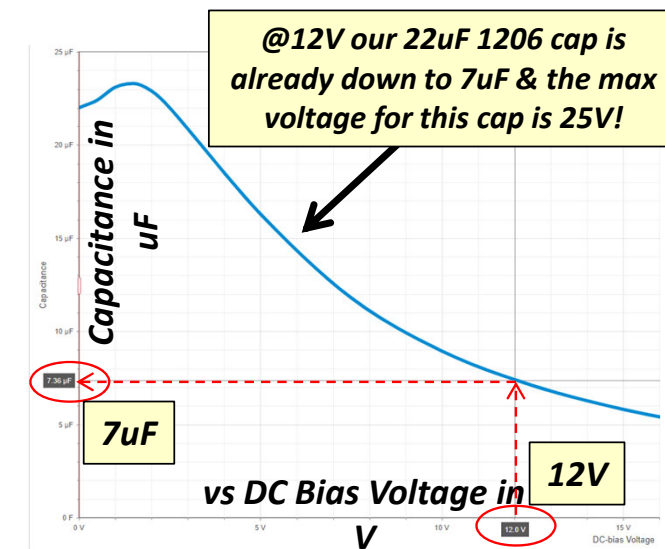
- In our case $\Delta V_{in_pk_pk} = 100\text{mV}$, $I_o = 2\text{A}$, $F_s = 200\text{kHz}$ and $D = 32\%$

$$C_{in} = \frac{2 \times 0.32 \times (1 - 0.32)}{200\text{kHz} \times \Delta V_{in_pk_pk}} = 22\mu\text{F}$$

- In other words if we add a $22\mu\text{F}$ ceramic capacitor as our C_{in} , then our $\Delta V_{in_pk_pk}$ should be around 100mV
 - Again we are assuming that the source impedance is high i.e. worst case
 - We almost always use multiple ceramics judiciously located around the board
 - This is the absolute minimum value, in practice you need 2 or 3 times this value due the DC Bias Loss
 - Please Google “Biricha YouTube DC Bias Loss” for our videos on this subject
- As mentioned earlier please also note that we often would like to add electrolytic bulk capacitance
 - This will may reduce the ripple even further → we will talk about thus shortly

Ceramic Cap Rated Voltage Selection → DC Bias Loss

- When we apply a DC voltage to a large value ceramic cap, its capacitance falls → this is called DC Bias Loss
 - This means that your 22uF capacitor will be much smaller than 22uF when you apply your DC voltage to it
 - e.g. a 22uF, 25V, 1206 package ceramic cap at 12V has a capacitance of only 7uF*!
 - Google “Biricha YouTube Bias loss” for our videos
- In order solve this issue:
 - Buy the one with a voltage rating at least 3 or 4 times your max DC voltage → above 10uF your choice will become limited
 - Use larger footprints → 1206, 1210 or bigger
 - Use Manufacturers’ tools (e.g. REDEXPERT from Würth) to work out the value of the capacitance after DC Bias Loss and install appropriate amount of capacitance *after* DC Bias Loss
 - Instead of one large high uF ceramic, use several smaller ones
 - If you can not estimate the DC Bias Loss then solder at least 3 or 4 x the amount of capacitance than you calculated i.e. if you calculated 22uF, start with 3 or 4 x 22uF!



***We now know everything to
select our ceramics***

What Do Ceramic Capacitor Temperature Stability Codes Mean?*

- Class II → high values and good for decoupling → Most Common for PSUs
 - e.g. X7R → X → -55°C, 7 → +125°C, R → ±15%
 - So if temp varies from -55 degrees to +125 degrees there will be a ± 15% change in the capacitance
 - All Zs and most Ys are out if your product is likely to freeze!
 - S, T, U & V are out for obvious reasons
- For most power supply applications X7R is a safe bet 😊

- There is also Class I which are very stable (temperature compensated) such as C0G
 - C0G means zero temperature drift!
 - But usually their values are small so often not suitable for PSUs
 - And ESR is large for power supplies → Just have a play on the manufacturer design tool

First letter indicates lower temperature limit	Middle number indicates upper temperature limit	Last letter indicates change of capacitance over the temperature range
X = -55 °C (-67 °F)	4 = +65 °C (+149 °F)	P = ±10%
Y = -30 °C (-22 °F)	5 = +85 °C (+185 °F)	R = ±15%
Z = +10 °C (+50 °F)	6 = +105 °C (+221 °F)	S = ±22%
	7 = +125 °C (+257 °F)	T = +22 to -33%
	8 = +150 °C (+302 °F)	U = +22 to -56%
	9 = +200 °C (+392 °F)	V = +22 to -82%

WDS Ceramic Capacitor Selection Example

- So for this PSU, we need $\sim 22\mu\text{F}$ and the voltage stress is $\sim 14\text{V}$

Frequency Response | Circuit | Power Loss Budget | Summary | Spice Simulation | **Input Capacitor**

Input Capacitor for Switching Ripple Reduction

Input voltage ripple (pk-pk) %

Input voltage ripple (pk-pk) mV

Ripple reduction capacitance (ceramic) μF

Input Ripple Summary

Overall input voltage ripple (pk-pk) mV

Overall input voltage ripple (rms) mV

Max. capacitor voltage V

Untitled - WDS - Licensed to Ali

File | Calculations | Edit | View | Help

Output Filter | **Specification** | Controller Design | Analog (Non-Isolated) | transformer | Semiconductors

Converter Specification

Topology:

Output voltage isolated from primary side:

Input Supply:

Maximum V

Nominal V

Minimum V

Output:

Maximum Current A

Voltage V

Output voltage ripple / overshoot:

We now know that $\sim 3 \times 22\mu\text{F}$ or $10\mu\text{F}$, 50V min, X7R caps with 1206, 1210 package or bigger would be a good ballpark

Let us select our capacitor using REDEXPERT

Ceramic Capacitor Selection from REDEXPERT

- REDEXPERT → Capacitors → Ceramics → Filter to show 22uF

- From previous slides we know that we need 22.99uF and the voltage stress on our capacitor is ~14V
 - But there will be some DC Bias Loss
 - At first glance 3x22uF would seem like a good choice but 50V versions are rare and expensive
 - Let us search from 10uF to 22F

WÜRTH ELEKTRONIK REDEXPERT Multilayer Ceramic Capacitors

Filters: C ≥ 10.0 µF

Order Code	Spec	Series	Description	Size	Ce...	Tol...	V _R	R _{ESR}	DF
885012214006	PDF	WCAP-CSGP	General Purpose	2220	X7R	10.0			25 %
885012214005	PDF	WCAP-CSGP	General Purpose	2220	X7R	10.0			25 %
885012214004	PDF	WCAP-CSGP	General Purpose	2220	X7R	10.0			25 %
885012214001	PDF	WCAP-CSGP	General Purpose	2220	X7R	10.0			25 %
885012209073	PDF	WCAP-CSGP	General Purpose	1210	X7R	10.0 µF ±10 %	50.0 V	> 5.00 MΩ	10 %
885012209028	PDF	WCAP-CSGP	General Purpose	1210	X7R	10.0 µF ±10 %	25.0 V	> 10.0 MΩ	5.0 %
885012209014	PDF	WCAP-CSGP	General Purpose	1210	X7R	10.0 µF ±10 %	16.0 V	> 100 MΩ	5.0 %
885012209005	PDF	WCAP-CSGP	General Purpose	1210	X7R	10.0 µF ±10 %	10.0 V	> 100 MΩ	5.0 %

Show parts where C is from: 10 µF to 22 µF

Reset Continue

Ceramic Capacitor Selection from REDEXPERT

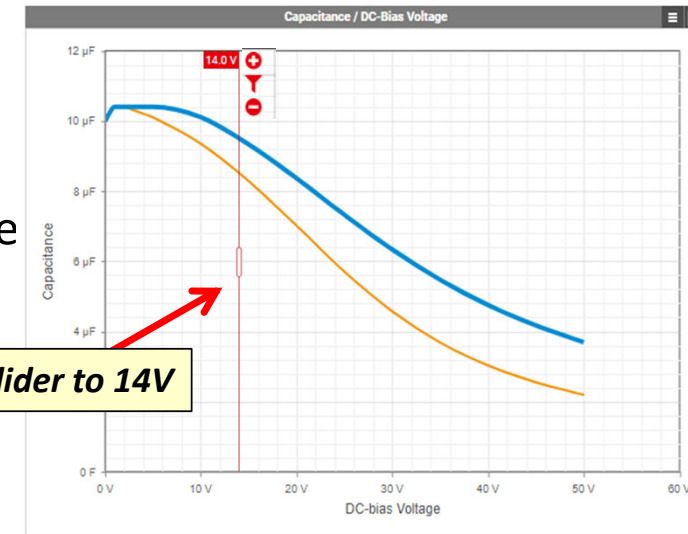
- Then let us further narrow down our choice to 50V and above capacitors → we have a total of 5 choices
 - They are all 10uF → no 22uF @ 50V
 - 1206 is only available in X5R → but we prefer X7R
 - So we select the 1210 and 2220, 50V versions only, and compare their DC Bias Loss @ our capacitor stress voltage of 14V

Filters:

	Order Code	Spec		Series	Description	Size	Ce...	C	Tole...	V _R
	885012108022			WCAP-CSGP	General Purpose	1206	X5R	10.0 µF	±10 %	50.0 V
☒	885012209073			WCAP-CSGP	General Purpose	1210	X7R	10.0 µF	±10 %	50.0 V
☐	885012214006			WCAP-CSGP	General Purpose	2220	X7R	10.0 µF	±10 %	100 V
☐	885012214001			WCAP-CSGP	General Purpose	2220	X7R	10.0 µF	±20 %	100 V
☒	885012214005			WCAP-CSGP	General Purpose	2220	X7R	10.0 µF	±10 %	50.0 V

Ceramic Capacitor Selection from REDEXPERT

- If we move the slider on the Capacitance vs DC-Bias graph our text based selection table also updates to display the DC Bias Loss at the position of the slider
 - In our case 14V DC



Filters: $10.0 \mu\text{F} \leq C \leq 22.0 \mu\text{F}$ $50.0 \text{ V} \leq V_R \leq 51.0 \text{ V}$

	Order Code	Spec	Series	Description	Size	Ce...	C	Tole...	V _R ...	R _{iso}	C(V _{DC-Bias}) @14.0 V
✓	885012214005		WCAP-CSGP	General Purpose	2220	X7R	10.0 µF	±10 %	50.0 V	> 10.0 MΩ	9.48 µF
✓	885012209073		WCAP-CSGP	General Purpose	1210	X7R	10.0 µF	±10 %	50.0 V	> 5.00 MΩ	8.49 µF
✓	885012108022		WCAP-CSGP	General Purpose	1206	X5R	10.0 µF	±10 %	50.0 V	> 5.00 MΩ	2.26 µF

Ceramic Capacitor Selection from REDEXPERT

Filters:

	Order Code	Spec	Series	Description	Size	Ce...	C	Tole...	V _R ...	R _{iso}	C(V _{DC} -Bias) @14.0 V
✓	885012214005		WCAP-CSGP	General Purpose	2220	X7R	10.0 μ F	$\pm 10\%$	50.0 V	> 10.0 M Ω	9.48 μ F
✓	885012209073		WCAP-CSGP	General Purpose	1210	X7R	10.0 μ F	$\pm 10\%$	50.0 V	> 5.00 M Ω	8.49 μ F
✓	885012108022		WCAP-CSGP	General Purpose	1206	X5R	10.0 μ F	$\pm 10\%$	50.0 V	> 5.00 M Ω	2.26 μ F

Input Capacitor - Switching Ripple

Input voltage ripple (pk-pk)

0.85 %

Input voltage ripple (pk-pk)

102 mV

Ripple reduction capacitance (ceramic)

22.99 μ F

Don't forget to update WDS!

- From REDEXPERT you can see that @14V DC
 - Our 10 μ F, 50V, 2220 package will have a capacitance of 9.48 μ F
 - Our 10 μ F, 50V, 1210 package will have a capacitance of 8.49 μ F
 - We excluded the 1206 package on the grounds that it was a X5R anyway, however it is interesting to note that @ 14V DC it would have a capacitance of only 2.25 μ F even though it is rated at 50V 😊
- To save space let us use 4 x 1220, 50V (part No 885012209073) for our design example! 4 x 8.49 $\approx$ 34 μ F
 - Actual voltage stress on the cap according to WDS $\sim$ 14V

Our ceramic capacitor selection is complete! 😊
But we must go back and update our Ceramic Cap values in WDS otherwise
Bulk Cap calculations will be wrong

Selecting Our Bulk Caps

- The size of our Bulk Capacitance (C_{bulk}) depends on two main factors
 - How much voltage dip we are allowed to have on V_{in} if we have a big load step
 - We must store enough energy in our Bulk caps so that after a known load step our V_{in} does not dip below a specified level
 - How much V_{in} can fall if our DC Bus is interrupted for a short period (thold)
 - We must store enough energy in our Bulk caps to be able to either tolerate this interruption or have enough time to shut things down gracefully
 - Note that for many smaller DC/DC power supplies, often thold is not even specified → we can then base our calculations on load step only
 - This is because at low voltages this calculation results in unfeasibly large capacitance requirements → hold up time most often used for AC-DC and high voltage DC-DC designs
 - Therefore we may have to calculate our Bulk Caps size twice (based on each requirement) and pick the worst case
 - If neither is specified then use your judgment 😊 e.g. don't let the voltage dip such that it violates circuit parameters such as Under Voltage Lockout UVLO of an IC, saturation limits of the inductor, etc.
- The time we have to hold V_{in} above a certain level after DC Bus interruptions (sometimes called Brown-Out) is called hold-up time or thold

Final Parameters before Bulk Cap Selection → Tolerance, Voltage Rating, Ripple Current Rating

- Tolerance: Most electrolytics/Polymer caps have a tolerance of $\pm 20\%$
 - So if you buy a 100uF capacitor, you may start with only 80uF to begin with!
 - Select one 20% bigger than the value that you need e.g. 120uF to make sure you have 100uF and then add another 20 to 30% tolerance to allow for aging/temperature, etc.
 - Voltage Rating
 - Electrolytics/Polymer do not suffer from DC Bias Loss so if you allow 20 to 30% tolerance of the maximum voltage rating you should be safe, e.g. if V_{in_max} is $\sim 14V$ a 16V or 25V cap will do
 - WDS calculates the maximum voltage stress on the capacitor but taking maximum value of the input voltage V_{in_max} is a safe bet
 - Very Important: Specified “Ripple” Current Rating ΔI_{Ripple_RMS}
 - The value on the datasheet is almost always the “*RMS value*” of the ripple current (not pk-pk!) with no DC component → sometimes called AC RMS
 - After we calculate our capacitance, we must calculate how much ripple current will pass through it (at F_s) and then buy a capacitor with a higher ripple current rating that we calculated
 - Of course WDS calculates all of these automatically, but for completeness we will provide the equations later in this presentation
- See Maniktala ISBN 978-0-12-386533-5, for much more details on aging/MTBF/reliability/etc. of capacitors

Impact of our Bulk Caps on the Overall Ripple Voltage

- In our case we used $\sim 34\mu\text{F}$ of ceramics to try to reduce our voltage ripple $\Delta V_{in_pk_pk}$ to less than 100mV
 - Ignoring DC Bias Loss for simplicity
- But our Bulk capacitance is in parallel with our ceramics and “should” reduce this ripple voltage even more
 - If we are using Aluminium *Electrolytics*, we can often ignore the impact of our Bulk capacitance on the ripple voltage
 - This is because at ripple frequency the capacitive impedance of the electrolytic is quite ineffective due to the high ESR
 - For Film/Aluminium *Polymer* caps you can not ignore this but then equations will get complicated and circular!
 - e.g. Work out the total impedance of a Bulk Cap in series with its ESR, then in parallel with a bunch of ceramics → then recalculate the ripple voltage based on the mix of ceramics and Bulk and readjust
 - Again WDS will calculate all of these for you, including the circular equations
 - Please see our videos about paralleling capacitors of differ types; Google “Biricha YouTube Paralleling Capacitors”

***We now know everything
to be able to start
calculating the size our
bulk caps***

Selecting Our Bulk Caps → Based on Load Transient

- From school level Physics we know that Power (P in Watts) is the rate of doing Work (WD or Energy in Joules) with respect to time (t in seconds), i.e.

$$P = \frac{WD}{t} \therefore WD = P \times t$$

$$WD = V \times I \times t \text{ (in Joules)}$$

- Let us continue with our original design example:
 - $V_{in_nom} = 12V$, $V_o = 3.3V$, $I_{o_max} = 2A$, $\eta \sim 100\%$ and $F_s = 200kHz$
 - Note that for simplicity we are assuming 100% efficiency; therefore, $P_{in} = P_{out} \rightarrow V_{in} \times I_{in} = V_o \times I_{out}$
- Let us now assume that after a load step from 50% to 100% for a duration of 10 switching cycles and we do not wish V_{in} to fall by more than 5% of our nominal input voltage V_{in_nom} during this period
 - In our case $V_{in_nom} = 12V$, 5% of 12V = 600mV
 - Stepping from 50% to 100% gives step current ΔI_o of 1A
 - $F_s = 200kHz \therefore$ Period $T = 1/F_s = 5\mu s$, so 10 periods = $50\mu s \rightarrow t = 50\mu s$
- From the equation above, the total “extra” energy that we need is given by

$$W = V_o \times \Delta I_o \times t = 3.3V \times 1A \times 50\mu s = 165\mu J$$

Selecting Our Bulk Caps → Based on Load Transient

- We need 165uJ & from school level Physics we also know that energy stored in a capacitor is:

$$WD = \frac{1}{2} CV^2$$

- This extra energy must NOT cause a voltage dip (ΔV_{in}) of more than 600mV as per our specification
 - Before load step → $V_{in} = 12V$
 - After load step → V_{in} must remain $\geq (V_{in} - \Delta V_{in}) = 12V - 600mV = 11.4V$
- Total energy that we must store in Bulk Caps must be bigger than 165uJ ∴

$$W = \frac{1}{2} CV_{in}^2 - \frac{CV_{in} - (V_{in} - \Delta V_{in})^2}{2} = \frac{1}{2} C \times 12^2 - \frac{1}{2} C \times 11.4^2 = 165\mu J$$

- In the equation we know all the variables → we can calculate our C

$$C_{bulk} \geq \frac{2 \times 165\mu J}{(12^2 - 11.4^2)} = 23.5\mu F$$

- Note: allow ~20 to 30% tolerance on capacitance & voltage rating
 - 33uF, 25V could be good choices for C_{bulk} 😊

All we have to do is make sure we buy a current with a ripple current rating bigger than our actual ripple current

How to Calculate the RMS Value of a Triangular Wave

- The waveform of most hard switched power supplies are choppy with fast edges, typically either triangular or trapezoidal in nature
 - We assume that our $\Delta V_{in_pk_pk}$ is in fact triangular $\rightarrow$ this is not too unreasonable as we have already added a lots of ceramics
- We also know from school that the RMS value of a triangular wave is its "peak" value (i.e. $pk-pk/2$) divided by the square root of 3, therefore:

$$\Delta V_{in_{RMS}} = \frac{\Delta V_{in_pk_pk}}{2} \times \frac{1}{\sqrt{3}}$$

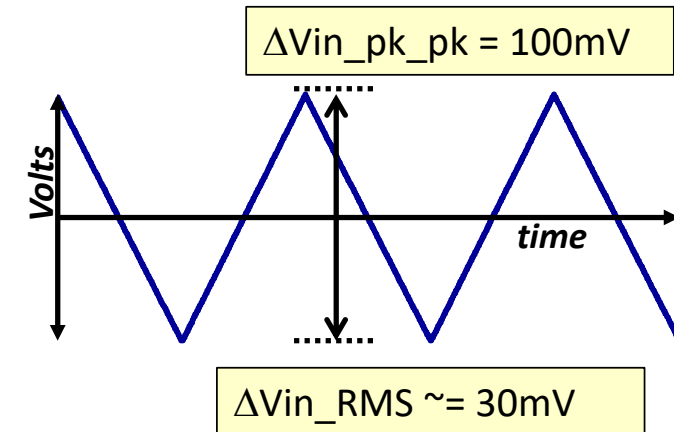
- In the previous slide we selected our ceramic capacitors so that $\Delta V_{in_pk_pk} \approx 100mV$, therefore our RMS value will be:

$$\Delta V_{in_{RMS}} = \frac{\Delta V_{in_pk_pk}}{2} \times \frac{1}{\sqrt{3}} \rightarrow 100mV / (2\sqrt{3}) \approx 30mV$$

- We can then calculate the total ripple current that will pass through our Bulk capacitor at our switching/ripple frequency

$$\Delta I_{Ripple_RMS} > \frac{\Delta V_{in_{RMS}}}{Z@Fs}$$

- We then buy a capacitor with a bigger current rating $\rightarrow$ Full design example shortly



Selecting Our Bulk Caps → Ripple Current Rating

- We now know ΔV_{in_RMS} across our bulk cap, we also know its impedance @ our ripple frequency (i.e. switching frequency)

$$Z = \sqrt{Z_c^2 + ESR^2} \rightarrow Z@Fs = \sqrt{\left(\frac{1}{2\pi Fs C}\right)^2 + ESR^2}$$

- So for a given capacitor, we know its capacitance and ESR + we know our switching frequency F_s
 - We can calculate RMS ripple current ΔI_{Ripple_RMS} and check against our caps datasheet
- We must buy a capacitor that has a ripple current rating bigger than:

$$\Delta I_{Ripple_RMS} > \frac{\Delta V_{in_RMS}}{Z@Fs}$$

- Finally please note that for electrolytics, we can often simplify this calculation by ignoring Z_c at F_s i.e. $Z@Fs = ESR$
 - This is because @ F_s , Z_c is usually significantly smaller than the ESR

- Example:
 - 100uF Electrolytic Cap with 100mΩ ESR
 - $F_s = 200kHz$
 - $\Delta V_{in_RMS} = 30mV$

$$\Delta I_{Ripple_RMS} \gg \frac{30mV}{100m\Omega} = 300mA$$

Selecting Our Bulk Caps → Ripple Current Rating

- Example 1: Electrolytic with large ESR

- 100uF Electrolytic Cap with 100mΩ ESR
- $F_s = 200\text{kHz}$
- $\Delta V_{in_RMS} = 30\text{mV}$
- Buy a capacitor with

$$\Delta I_{Ripple_RMS} \gg \frac{30\text{mV}}{100\text{m}\Omega} > 300\text{mA}$$

- Example 2: As above but this time Polymer cap with small ESR

- 100uF Aluminium Polymer with 5mΩ ESR

$$Z@FS = \sqrt{\left(\frac{1}{2\pi \cdot 200\text{kHz} \cdot 100\mu\text{F}}\right)^2 + 5\text{m}\Omega^2} = 9.5\text{m}\Omega$$

$$\Delta I_{Ripple_RMS} \gg \frac{30\text{mV}}{9.4\text{m}\Omega} > 3.20\text{A}$$

- Don't let the large ripple current of the Polymer cap put you off, they have much high ripple current ratings

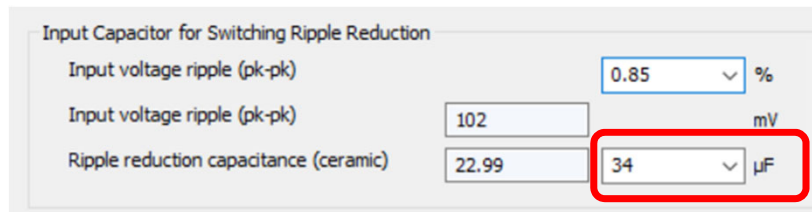
- Due to their low ESR ☺

Let us now calculate and select our capacitor

We have all the equations for completeness but for simplicity we will use WDS

Bulk Capacitor Calculation Based on Transient Requirement Complete Example

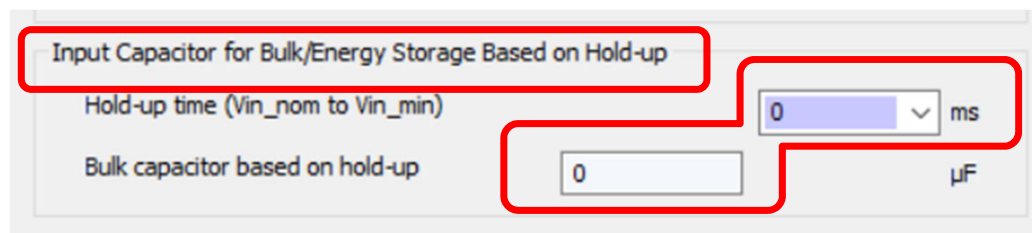
- Firstly please remember that we have 34uF worth of ceramics
 - You must update WDS or all the rest of the calculations will be wrong



Input Capacitor for Switching Ripple Reduction

Input voltage ripple (pk-pk)	0.85	%
Input voltage ripple (pk-pk)	102	mV
Ripple reduction capacitance (ceramic)	22.99	34 μF

- For simplicity let us assume that for now we do not have any hold up time specification, therefore we base our selections on transient requirements only
 - We will provide a hold up time example later on in the notes



Input Capacitor for Bulk/Energy Storage Based on Hold-up

Hold-up time (Vin_nom to Vin_min)	0	ms
Bulk capacitor based on hold-up	0	μF

WDS Bulk Capacitor Calculation Complete Example

Calculated value

This is the ripple current through the Bulk Cap, you must not buy a capacitor with a lower ripple current rating that is shown here. The value of the ESR must be correct or this value will be wrong!

Input Capacitor for Bulk/Energy Storage Based on Load Transient

Input cap volt. drop during output transient: 5 %

Input cap volt. drop during output transient: 600 mV

Output transient as percentage of max load: 50 %

Recovery time to recharge input cap: 50 μ s

Bulk capacitor based on transient: 23.5 μ F

Bulk/Energy Storage Capacitor Selection

Bulk storage capacitor: 23.5 33 μ F

Bulk storage capacitor impedance (at Fs): 260.61 260.61 m Ω

Bulk storage capacitor ripple current (pk-pk): 260.35 mA

Bulk storage capacitor ripple current (rms): 75.16 mA

Input Ripple Summary

Ripple reduction capacitance (ceramic)	34	μ F
Bulk storage capacitor	33	μ F
Bulk storage capacitor ripple current (rms)	75.16	mA
Overall input voltage ripple (pk-pk)	68.14	mV
Overall input voltage ripple (ac rms)	19.67	mV
Max. capacitor voltage	14.03	V

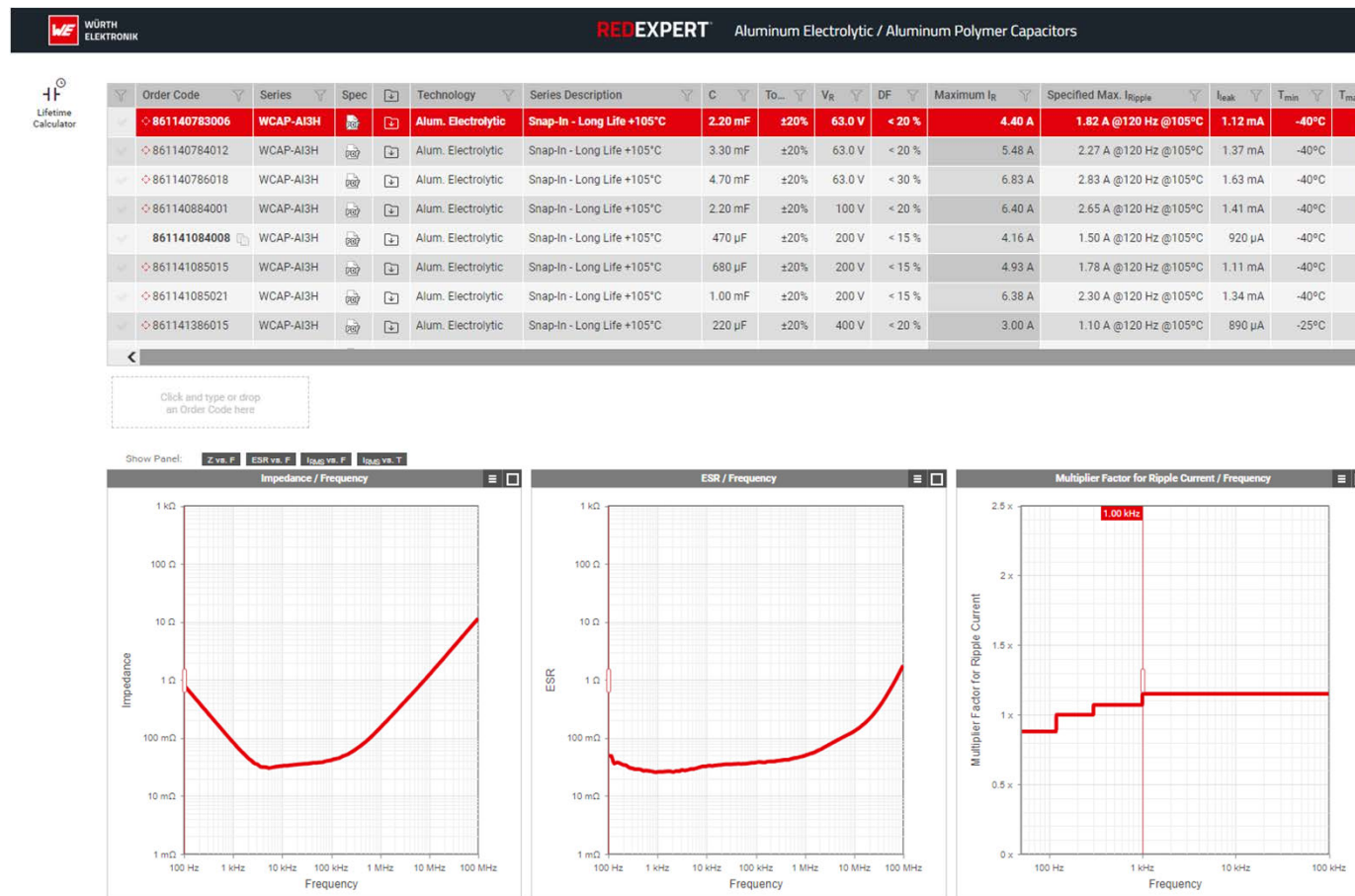
User selected value

Very important: this is only an estimate! you must type in the correct value of the impedance of your chosen cap either by measuring or datasheet or online tool

Summary of all the data that you need to select both types of your capacitances

Bulk Capacitor Selection Complete Example

• REDEXPERT Capacitor Selection Tool



- Based on our pervious calculation, we are looking for a 33uF, 25V Aluminium Electrolytic Capacitor
- We must make sure that we do not violate the ripple current rating
 - We need either the ESR at F_s or total Z @ F_s
 - WDS will then calculate everything that we need, but for completeness we will work it out by hand also later

Bulk Capacitor Selection Complete Example

- Step 1: Click on Capacitance and filter to show only 33uF

REDEXPERT Aluminum Electrolytic / Aluminum Polymer Capa

Description	C...	To...	V _R	DF	Maximum I _R
- In - General Purpose +10...	33.0				655 mA
- Long Life +105°C	33.0				41.6 mA
- Long Life +105°C	33.0				44.2 mA
- Long Life +105°C	33.0				57.2 mA
- Long Life +105°C	33.0 µF	±20%	25.0 V	< 16 %	62.4 mA
- Long Life +105°C	33.0 µF	±20%	35.0 V	< 13 %	74.1 mA
- Long Life +105°C	33.0 µF	±20%	50.0 V	< 12 %	104 mA
- Low Impedance +105°C	33.0 µF	±20%	6.30 V	< 24 %	73.0 mA

Show parts where

C is from:

33 µF to 33

Bulk Capacitor Selection Complete Example

- Step 2: Filter to show rated voltage of 25V only

	C	To...	V _R ...	DF	Z@200 kHz	Maximum I _R	Specified Max. I _{Ripple}	I _{leak}
	33.0 µF	±20%	16.0 V			63.7 mA	49.0 mA @120 Hz @85°C	1.06 µA
	33.0 µF	±20%	16.0 V			63.7 mA	49.0 mA @120 Hz @85°C	5.28 µA
	33.0 µF	±20%	16.0 V			92.0 mA	115 mA @100 kHz @105°C	5.28 µA
25°C	33.0 µF	±20%	16.0 V			124 mA	91.0 mA @120 Hz @125°C	5.28 µA
5°C	33.0 µF	±20%	16.0 V	< 15 %	704 mΩ	87.0 mA	64.0 mA @120 Hz @105°C	5.28 µA
°C	33.0 µF	±20%	16.0 V	< 16 %	1.06 Ω	114 mA	84.0 mA @120 Hz @85°C	5.28 µA
+10...	33.0 µF	±20%	16.0 V	< 16 %	991 mΩ	120 mA	144 mA @100 kHz @105°C	5.28 µA
	33.0 µF	±20%	16.0 V	< 8.0 %	23.5 mΩ	660 mA	2.20 A @100 kHz @105°C	400 µA

Show parts where

V_R is from:

25 V to 25 V

Reset Continue

Bulk Capacitor Selection Complete Example

- Step 3: Filter other requirements → SMT, 105°C, no polymer, etc.

max	Lifetime	Asse...	Width	Height	Length
85.0°C	1000.0 h	SMT			
85.0°C	2000.0 h	SMT			
105°C	2000.0 h	SMT			
105°C	5000.0 h	THT			
125°C	1000.0 h	THT			
105°C	2000.0 h	THT		11.0 mm	
85.0°C	2000.0 h	THT		11.0 mm	
105°C	5000.0 h	THT		11.0 mm	

Show parts where

Assembling Technology is equal to:

DF	Maximum I _R	Specified Max. I _{Ripple}	I _{leak}	T _{min}	T _{max}	Lifetime	Asse...
V	< 22 %	57.2 mA	44.0 mA @120 Hz @105°C	5.28 μA	-40°C	105	
V	< 16 %	157 mA	185 mA @100 kHz @105°C	5.28 μA	-55°C	105	
V	< 16 %	205 mA	230 mA @100 kHz @105°C	5.28 μA	-55°C	105	
V	< 8.0 %	660 mA	2.20 A @100 kHz @105°C	400 μA	-55°C	105	

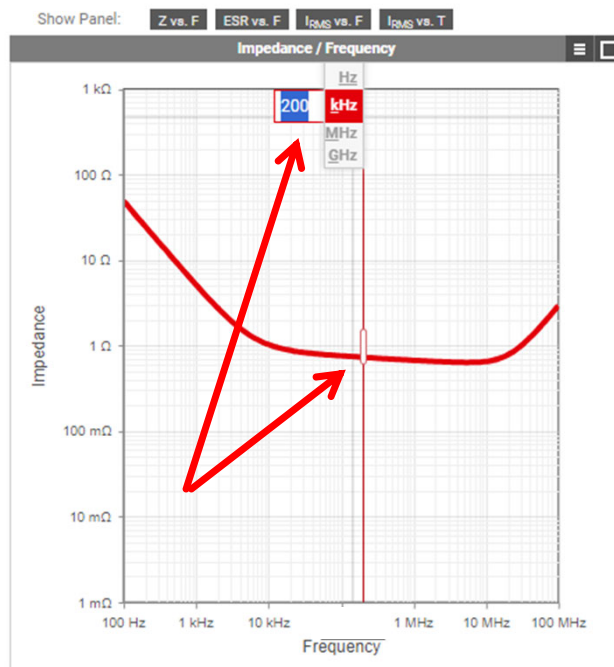
Show parts where

T_{max} is equal to:

We will normally have a choice of only 3 or 4 which so far meet our criteria 😊

Bulk Capacitor Selection Complete Example

- Step 4: On the Impedance vs Frequency plot, set the slider to your switching frequency → in our case 200kHz
- This will in turn display all the impedances of all the capacitors at 200kHz only in the Z@ columns in the table

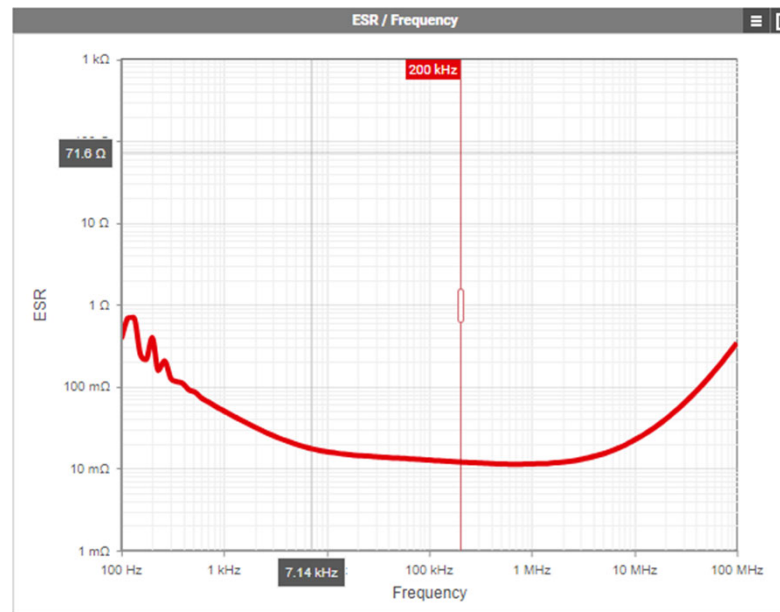


Z@200 kHz...	Maximum I_R	Specified Max. I_{Ripple}	I_{leak}
1.16 Ω	142 mA	160 mA @100 kHz @105°C	8.2
704 m Ω	62.4 mA	48.0 mA @120 Hz @105°C	8.2
524 m Ω	205 mA	230 mA @100 kHz @105°C	8.2
37.2 m Ω	270 mA	900 mA @100 kHz @105°C	8.2
29.5 m Ω	801 mA	2.67 A @100 kHz @105°C	60

Bulk Capacitor Selection Complete Example

- Though not directly needed for WDS, you can also display the ESR at a certain frequency on the table by moving the ESR vs F slider

ESR @200 kHz	Maximum I_R	Specified Max. I_{Ripple}
12.0 mΩ	660 mA	2.20 A @100 kHz @105°C
609 mΩ	205 mA	230 mA @100 kHz @105°C
468 mΩ	157 mA	185 mA @100 kHz @105°C
727 mΩ	57.2 mA	44.0 mA @120 Hz @105°C



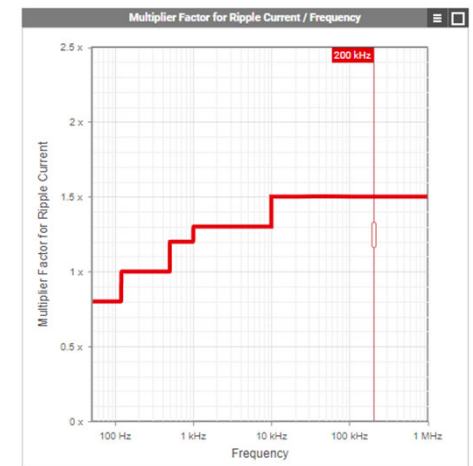
Bulk Capacitor Selection Complete Example

- Step 5: Get REDEXPERT to sort them based on their ripple currents
 - Important/common error → for PSU Input Cap applications we are looking for “Ripple Current I_{Ripple} ” NOT “Rated Current Maximum I_R ”

- Note that there may be a multiplier factor graph but at high frequencies we can have higher ripple currents so this just an added bonus
- On many modern capacitors, ripple current rating is usually given at 100kHz with a multiplier factor of 1 for higher frequencies

We need this value

	C	To...	V _R	DF	Z@200 kHz	Maximum I _R	pecified Max. I _{Ripple}	I _l
	33.0 µF	±20%	25.0 V	< 8.0 %	29.5 mΩ	801 mA	2.67 A @100 kHz @105°C	
C	33.0 µF	±20%	25.0 V	< 14 %	37.2 mΩ	270 mA	900 mA @100 kHz @105°C	
10...	33.0 µF	±20%	25.0 V	< 14 %	524 mΩ	205 mA	230 mA @100 kHz @105°C	
:	33.0 µF	±20%	25.0 V	< 14 %	1.16 Ω	142 mA	160 mA @100 kHz @105°C	
	33.0 µF	±20%	25.0 V	< 16 %	704 mΩ	62.4 mA	48.0 mA @120 Hz @105°C	



Bulk Capacitor Selection Complete Example

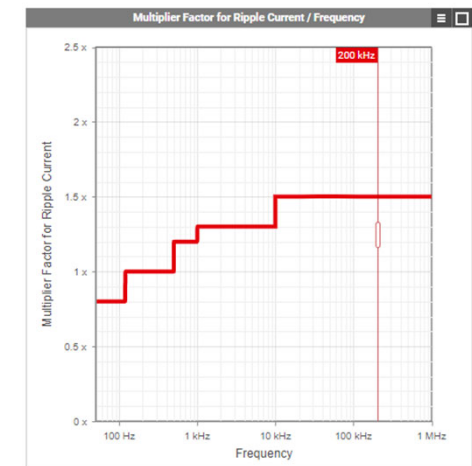
Order Code	Series	Spec	Technology	Series Description	C	To...	V _R	DF	Z@200 kHz	ESR @200 kHz	Maximum I _R	Specified Max. I _{ripple}
875105545004	WCAP-PSLP		Alum. Polymer	SMT - Low Profile	33.0 μ F	$\pm 20\%$	25.0 V	< 8.0 %	29.5 mΩ	18.7 mΩ	801 mA	2.67 A @100 kHz @105°C
875575559001	WCAP-HSG5		Hybrid Polymer	SMT - General Purpose +105°C	33.0 μ F	$\pm 20\%$	25.0 V	< 14 %	37.2 m Ω	25.1 m Ω	270 mA	900 mA @100 kHz @105°C
865060443003	WCAP-ASLL		Alum. Electrolytic	SMT - Low Imp. & Long Life +10...	33.0 μ F	$\pm 20\%$	25.0 V	< 14 %	524 m Ω	517 m Ω	205 mA	230 mA @100 kHz @105°C
865080442006	WCAP-ASLI		Alum. Electrolytic	SMT - Low Impedance +105°C	33.0 μ F	$\pm 20\%$	25.0 V	< 14 %	1.16 Ω	1.15 Ω	142 mA	160 mA @100 kHz @105°C
865230443004	WCAP-AS5H		Alum. Electrolytic	SMT - Long Life +105°C	33.0 μ F	$\pm 20\%$	25.0 V	< 16 %	704 m Ω	699 m Ω	62.4 mA	48.0 mA @120 Hz @105°C

- You can see that we have a choice of 5
- The 1st cap (part # x45004) is Aluminium *Polymer* (as opposed to Aluminium *Electrolytic*)
 - It has the highest ripple current rating because of the *significantly* lower ESR → have a quick look at the ESR of the electrolytic variants on the same table!
 - It has the lowest impedance at our switching frequency → again due to the low ESR
 - but it is likely that it will be more expensive → let us disregard this one unless we can not find a cheaper electrolytic capacitor
- Let us also exclude our Hybrid Polymer (x59001) and just stick with electrolytics
- The last capacitor (x43004) has the lowest ripple current rating and is this specified at 120Hz for line frequency application
 - Let us disregard this also as we already have 2 better ones!

Bulk Capacitor Selection Complete Example

Order Code	Series	Spec		Technology	Series Description	C	To...	V _R	DF	Z@200 kHz	ESR @200 kHz	Maximum I _R	Specified Max. I _{Ripple}
865060443003	WCAP-ASLL			Alum. Electrolytic	SMT - Low Imp. & Long Life +10...	33.0 μF	±20%	25.0 V	< 14 %	524 mΩ	517 mΩ	205 mA	230 mA @100 kHz @105°C
865080442006	WCAP-ASLI			Alum. Electrolytic	SMT - Low Impedance +105°C	33.0 μF	±20%	25.0 V	< 14 %	1.16 Ω	1.15 Ω	142 mA	160 mA @100 kHz @105°C

- We are left with only 2 choices (x43003 & x42006)
 - Z@Fs are 524m Ω and 1.16m Ω respectively
 - Ripple current ratings are 230mA and 160mA respectively*
- The question is: “At F_s , will our RMS ripple voltage across either of these caps cause a ripple current of higher than what is rated?”
 - We have Z@Fs, We have ΔV_{in_RMS} $\therefore$ we can calculate our ripple current as shown in previous examples
 - Unlike other technologies, For electrolytics, we can often use ESR if we don’t have Z@Fs $\rightarrow$ just compare ESR@Fs with Z@Fs!
- Equations and numerical examples have already been given, so we will just use WDS
 - If you type in Z@Fs into WDS, it will calculate what ripple current will flow $\rightarrow$ this calculated ripple current must be less than the rated ripple current shown here



* We are ignoring the multiplier factor for simplicity but it is x1 anyway!

Bulk Capacitor Selection Complete Example

Order Code	Series	Spec		Technology	Series Description	C	To...	V _R	DF	Z@200 kHz	ESR @200 kHz	Maximum I _R	Specified Max. I _{Ripple}
865060443003	WCAP-ASLL			Alum. Electrolytic	SMT - Low Imp. & Long Life +10...	33.0 μF	±20%	25.0 V	< 14 %	524 mΩ	517 mΩ	205 mA	230 mA @100 kHz @105°C
865080442006	WCAP-ASLI			Alum. Electrolytic	SMT - Low Impedance +105°C	33.0 μF	±20%	25.0 V	< 14 %	1.16 Ω	1.15 Ω	142 mA	160 mA @100 kHz @105°C

- Capacitor part number x43003

- Type in Z@Fs of 1160m Ω * into WDS and WDS will calculate the ripple current that will flow (i.e. 17.15mA)
- 17.15mA is less than the specified rated ripple current from the manufacturer (i.e. 160mA) $\rightarrow$ this could be a suitable choice but the ESR is quite big

Our Bulk Cap selection is complete 😊
But what if we had hold up time requirements ?

Bulk/Energy Storage Capacitor Selection

Bulk storage capacitor	23.5	33	μ F
Bulk storage capacitor impedance (at Fs)	260.61	1160	m Ω
Bulk storage capacitor ripple current (pk-pk)	59.4		mA
Bulk storage capacitor ripple current (rms)	17.15		mA

- Repeat this for X42006 $\rightarrow$ is this part suitable ?

* We are ignoring the multiplier factor for simplicity but it is x1 anyway!

Selecting Our Bulk Caps → Based on Holdup Time

- How much energy do we have to store in C_{bulk} so that after 10ms, our voltage does not fall below 10V? i.e. How big does our capacitor need to be?
 - The procedure is almost exactly the same

- We need to store this much energy:

$$W = \frac{1}{2} C V_{in(nom)}^2 - \frac{1}{2} C V_{in(min)}^2$$

- We know that WD = Pxt in our case 6.6Wx10ms = 0.066J
- We know V_{in_nom} = 12V and V_{in_min} = 10V so

$$0.066J = \frac{1}{2} C 12^2 - \frac{1}{2} C 10^2 \rightarrow C_{bulk} = \frac{0.066 \times 2}{12^2 - 10^2} = 3000\mu F$$

- C_{bulk} based on holdup time is much bigger than C_{bulk} based on load step → We select the worst case
 - Of course WDS makes this calculation for you

As expected 3000uF is far too large, that is because Vin is low (WD ∝ V²)

Input Capacitor for Bulk/Energy Storage Based on Hold-up

Hold-up time (V_{in_nom} to V_{in_min}) ms

Bulk capacitor based on hold-up μF

- Hold up time is rarely used for low voltage designs as it gives unfeasibly large results but let us try anyway 😊
- V_{in_max} = 14V, V_{in_nom} = 12V, V_{in_min} = 10V, Input Power P_{in} = 6.6W, η ≈ 100% and F_s = 200kHz
- Let us now assume that after a brownout time of t_{hold} = 10ms our V_{in} can not drop below 10V due to some undervoltage lockout requirement

Our Bulk Cap selection based on hold time is also is complete 😊

Summary

- In this presentation we discussed why we need input capacitance and what types we need
 - Ceramics and Electrolytics
- We showed exactly how you calculate the size of the ceramics
 - We gave all the necessary equations + a numerical example but WDS calculates everything for you anyway
- We showed exactly how you calculate the amount of bulk capacitance that you need based on either transient requirements or hold up time
- We gave all the necessary equations + hand calculated a numerical
- We used WDS and REDEXPERT to very quickly select our capacitor

***End of presentation
Thank you for your time***